Predictive latent dynamics for parametric vortex-gust airfoil interactions at Re = 5000

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Abstract

We ask whether a predictive latent objective yields a reduced-order state for parametric vortex-gust airfoil interactions that forecasts forward in time better than a reconstructive one. The data are our own direct numerical simulations (spectral-element solver SOD2D) of a NACA 0012 at angle of attack $\alpha = 14°$ and chord Reynolds number Re = 5000, perturbed by a discrete Taylor vortex parametrised by gust strength G, core diameter D, and wall-normal offset Y, spanning 84 cases (226 training, 28 validation, 42 in-distribution test, and 24 out-of-distribution extrapolation encounters). Three encoders are compared at matched latent dimension, a joint-embedding predictive architecture (JEPA), a reconstructive autoencoder after Fukami & Taira, and a proper orthogonal decomposition basis, under one shared autoregressive predictor, training recipe, conditioning on (G, D, Y), and probe family, so that only the encoder varies. We judge each state not by reconstruction but by *forward physical closure* of six observables spanning body forces, integrated impulse, and wake diagnostics, sixteen frames after gust impact. The predictive latent leads on the wake observables: on held-out encounters its representational wake-enstrophy R^2 is positive (0.75) where the reconstructive and linear baselines are negative, and a per-encounter paired test, which cancels the shared encounter difficulty, gives it the smaller wake-enstrophy error on 31 of 42 encounters representationally and 27 of 42 under forecast (one-sided sign $p = 1.4 \times 10^{-3}$ and 4.4×10^{-2}). The mechanism is latent drift: under rollout the reconstructive latent leaves its training manifold by an order of magnitude while the predictive and linear latents stay within it. A conditioning-only floor and a matched-architecture, matched-supervision 2×2 control attribute the gain to the predictive objective trained with wake supervision, not to the architecture. The predictor is the latent dynamics function a model-based controller plans against, establishing the predictive objective as a route to forecastable reduced-order states for unsteady aerodynamics.

Key words: vortex shedding, low-dimensional models, machine learning

1 Introduction

Small uncrewed and micro air vehicles increasingly operate in airspace where the gust velocity is comparable to or larger than their flight speed, such as urban canyons, mountainous terrain, and the wakes of buildings and ships [11, 16]. In this regime the gust ratio G, the ratio of the characteristic gust velocity to the free-stream velocity, exceeds unity, and the interaction between a discrete vortex gust and a lifting surface produces large, fast transients in the aerodynamic loads that conventional gust models do not capture [6, 7]. The response is governed by the timing of the vortex impact relative to the airfoil's own shedding, by the wall-normal offset of the vortex core, and by the gust strength and core size. Even at the moderate chord Reynolds number Re = 5000, the post-stall flow at $\alpha = 14°$ is massively separated, and the gust-induced disturbance interacts nonlinearly with the natural shedding cycle [7]. Resolving the full parametric envelope by simulation is expensive, so reduced-order models (ROMs) of the encounter-to-encounter and case-to-case variability are an active line of work, and a reduced model is also the object a controller would plan against.

Recent ROMs for separated and extreme-aerodynamic flows share a common ingredient: a reconstruction-trained autoencoder whose bottleneck is shaped by an auxiliary aerodynamic observable. Fukami & Taira [8] trained a convolutional autoencoder jointly with a lift-prediction head and found that the auxiliary loss reduced the intrinsic dimension of the discovered manifold for extreme vortex-gust flows from roughly ten to three. The same observable-augmentation idea was extended across multi-source turbulent data [9], and was used by Solera-Rico et al. [22] with a β-variational autoencoder and a transformer ROM on a vortex-gust airfoil dataset of the same family considered here. Across this lineage the reconstruction term anchors the latent geometry to the instantaneous flow field, and an observable-prediction term orients it along aerodynamically relevant directions. A recurring theme of the geometric-analysis literature, however, is that the latent coordinates of an autoencoder are unique only up to a diffeomorphism, so reconstruction leaves the geometry of the latent largely unconstrained, which complicates any downstream task that relies on that geometry, including dynamics, interpolation, and comparison [23, 12].

A complementary route to a reduced state is to drop the reconstruction term altogether and to train the encoder and a predictor end to end on latent-space forecasting, with an explicit anti-collapse regulariser in place of the decoder. This is the joint-embedding predictive architecture (JEPA) [13]: an encoder maps the input to a representation, a predictor advances that representation in time, and the loss never reconstructs the field. Two consequences follow. The encoder is free to discard detail that is not predictive of the future, and the predictor cannot exploit pixel-level shortcuts because no field-space signal reaches it. The image and video instantiations [1, 2] and the recent from-pixels variants that remove the moving-target encoder in favour of a characteristic-function regulariser [14, 3] or a variance-covariance objective [20, 4] have so far been evaluated on gridworld and toy visual domains. Their behaviour on a low-intrinsic-dimension physics dataset, where the data sit on a roughly five- to ten-dimensional manifold, is uncharacterised.

Reconstruction-trained nonlinear encoders compress aggressively but yield latent dynamics that are harder to forecast than an energy-optimal linear basis, a compactness-versus-forecast tradeoff documented for controlled wake flows in a companion study under review [21]. We show this tradeoff is a consequence of the reconstruction objective: a predictive latent recovers nonlinear compactness without surrendering forecast stability.

This places the present model within the action-conditioned world-model framework articulated by [13]: a system is observed through an encoder into an abstract state, an intervention is applied, and a predictor forecasts the outcome of that intervention in the representation space rather than in the space of the raw observation. The defining property of such a model is that it does not reconstruct every detail of the outcome; it predicts only in the abstract space. The vortex gust is the intervention. A discrete vortex of strength G, size D, and offset Y is released into the post-stall wake, and the encounter is the outcome of that intervention; the encoder maps the vorticity field to an unconditional latent state, and the predictor advances that state conditioned on the gust parameters. We use the gust as an observed, exogenous intervention rather than an agent's chosen action, so the model is the interventional world model a controller would use, not yet a controller; and because we condition on the true generative parameters but do not perform counterfactual identification, we keep to the interventional reading and do not claim a full causal model. The reconstructive and linear baselines we compare against are, in this framing, the objects the world-model view argues against: a reconstruction-trained autoencoder is pushed toward reproducing every detail of the field, and a proper orthogonal decomposition is an energy-optimal reconstruction basis, neither of which is constrained to make the latent state forward-predictable. The drift mechanism of §4.3 is the empirical consequence.

The question this paper addresses is therefore not which encoder reconstructs the flow best, but which produces a reduced state that remains useful when propagated forward in time. We treat this as a fluid-mechanics question with a physical answer: a reduced state for gust-airfoil forecasting should be judged by whether its forward evolution keeps physically meaningful observables close to the simulation, and by whether the latent trajectory it generates stays on the manifold the model was trained on. We make three contributions.

First, we set up a controlled comparison in which a predictive encoder (JEPA), a reconstructive encoder in the Fukami lineage, and a POD basis are evaluated at matched latent dimension under a single shared autoregressive predictor, training recipe, parametric conditioning on $\mathbf{c} = (G, D, Y)$, and probe family, so that forecast quality is attributable to the encoder alone (§3). The figure of merit is forward physical closure of the rollout to six observables at horizon $H = 16$ (§4.1). On held-out encounters the predictive latent closes the spatially distributed wake observables markedly better than the reconstructive and linear baselines, while the linear basis remains competitive on the integrated impulse.

Second, we identify the mechanism. A latent-drift diagnostic shows that the reconstructive rollout leaves its training manifold by an order of magnitude at $H = 16$, whereas the predictive and linear rollouts remain within it (§4.3). We strengthen this from a single distance into a geometric and topological statement: the predictive latent organises the encounter as a closed cycle whose topology survives rollout, and its latent metric tracks the physical transport geometry of the flow more faithfully than the reconstructive latent does.

Third, we are explicit about what the comparison does and does not isolate. The predictive encoder differs from the reconstructive baseline not only in its objective but also in its architecture and in its auxiliary observable supervision. We therefore phrase the result as a property of the *predictive latent family*, report a conditioning-only floor that rules out the shared parametric conditioning as the explanation (§4.1), and set out the matched-architecture and matched-auxiliary controls required to attribute the gain to the predictive objective in isolation (§4.5). We close by relating the predictor to the latent dynamics function of model-based control, including a deployment pilot that does not yet meet its tolerances and is reported as a limitation rather than a result (§5).

2 Flow configuration and data

2.1 Vortex gust-airfoil interaction

We study a NACA 0012 airfoil at angle of attack $\alpha = 14°$ and chord Reynolds number $\mathrm{Re} = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The data analysed here are our own direct numerical simulations of this configuration, computed with the GPU-enabled spectral-element solver SOD2D [10] at low Mach number, so the flow is effectively incompressible. The configuration and its physical characterisation follow Fukami, Smith & Taira [7], which is the reference for the physics, not the source of the data. The gust is parametrised by three scalars: the signed gust ratio G, the core diameter D, and the wall-normal offset Y/c of the vortex centre relative to the leading edge. The training envelope spans $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$ chords, and $Y/c \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords and set to zero when the vortex centre reaches the leading edge.

Without a gust, the $\alpha = 14°$ wake at this Reynolds number is massively separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages [19, 7, 18]. The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The six observables defined in §2.2 are quantitative summaries of exactly this sequence.

A consequence of the source characterisation [7] is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the quantity the encoder observes no longer determines the true dynamics. We therefore treat $|G| = 4$ as a physical observability boundary rather than as a claim of parametric generalisation.

2.2 Data, fields, and observables

Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05\, t/c$, one encounter per gust release. The simulations use a spectral-element discretisation of polynomial order $p = 4$, and the fields are cached on a structured grid of $192 \times 96 \times 32$ points spanning $x/c \in [-1.5, 4.5]$, $y/c \in [-1.5, 1.5]$, and a spanwise extent of approximately one chord. The remaining solver-resolution details, namely the element count, the near-wall grid spacing relative to the local small scale, and the Mach number, will be supplied by the simulation collaborators before submission. The dataset comprises 84 cases partitioned into training, validation, an in-distribution test set (test_b), and an out-of-distribution test set (test_c). The in-distribution test set is stratified to span the five gust strengths, the three core diameters, and the two source groups (Figure 1); test_c is the $|G| = 4$ extrapolation set. At the horizon $H = 16$ used throughout, the partition contains 226 training, 28 validation, 42 test_b, and 24 test_c encounters. The undisturbed reference case is retained in training. The partition is fixed once and used identically by every model and baseline.

For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$, the span-averaged wall pressure $p_{\text{wall}}(192)$, and the lift and drag coefficients C_L, C_D. The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter high-percentile clip caps spikes, and the field is scaled by the training standard deviation with no mean shift, preserving the vorticity antisymmetry. All representation and probe losses are computed in this normalised space; unnormalised values are used only for physical-units metrics and figures.

We evaluate forward closure against six physical observables that together span body-force, integrated-flow, and spatially distributed wake diagnostics. The lift and drag coefficients C_L and C_D are the body forces. The wall-normal impulse I_y is an integrated-flow quantity coupled to the wake-vortex dynamics. The wake enstrophy and the positive and negative circulation are spatially distributed wake observables: they integrate (signed) vorticity over the wake region and are the most demanding to forecast, because they are sensitive to the redistribution of vorticity that the LEV roll-up and shedding produce. The choice is deliberate: the wake observables can change sharply while the scalar forces change little, so they are the part of the encounter most likely to separate a representation that has captured the gust-vortex physics from one that has only captured its force signature.

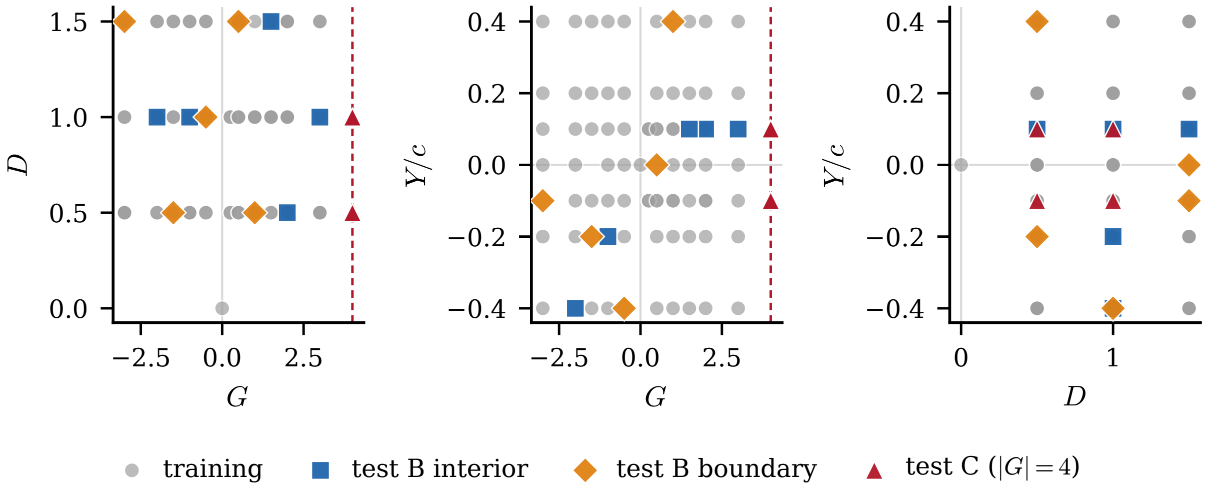

Figure 1: Parameter-space sampling of the 84 cases in the (G, D, Y) gust envelope, shown as three planar projections and coloured by split: training (226 encounters), the stratified in-distribution test set (test_b, 42 encounters, split into interior and boundary tiers), and the out-of-distribution extrapolation set (test_c, 24 encounters). The dashed line marks the $|G| = 4$ extrapolation boundary, and training mass concentrates near $Y/c = 0$, which is why the wall-normal offset is the least well resolved parameter (§4.4).

3 Predictive latent representation and evaluation protocol

3.1 Encoder, predictor, and the conditioning split

The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at $d \in \{32, 64\}$. It is a hybrid network: three convolutional downsampling stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. The encoder is *unconditional* by design: it sees only the flow field, not the gust parameters $\mathbf{c} = (G, D, Y)$. This is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5).

The predictor P_ϕ advances the latent in time. It is a six-layer autoregressive transformer with rotary positional embeddings and a causal mask, conditioned on $\mathbf{c}$ through adaptive layer-norm modulation at every block; its internal structure is given in Appendix A. The predictor is the only stage that sees $\mathbf{c}$. Figure 2 shows the encoder and predictor, and Figure 3 contrasts the predictive training signal with the reconstructive one: the reconstructive autoencoder passes the field at time t through an encoder and a decoder and incurs a field-space error, whereas the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss.

3.2 Training objective and the auxiliary observable heads

The predictive encoder and predictor are trained jointly on a one-step latent-prediction term, a scheduled-sampling rollout term that exposes the predictor to its own errors, and an anti-collapse regulariser that prevents the encoder from collapsing to a constant. We use the characteristic-function regulariser of [3, 14], which matches the latent distribution to an isotropic Gaussian through a small number of random projections, with an automatic fall-back to a variance-covariance objective [4] should the participation ratio of the latent indicate collapse. Its configuration is given in Appendix A.

Two auxiliary heads read aerodynamic observables off the encoder output: a lift head that maps $\mathbf{z}_t$ to $C_L(t)$, and a wake head that maps $\mathbf{z}_t$ to an 80-dimensional signed wake spectrum, trained with small weights jointly with the representation. These heads are adopted from the observable-augmentation lineage [8, 9]. They are also, however, the reason the present predictive encoder is not a clean isolation of the training objective: the reconstructive baseline below carries a lift head but not a wake head, and the two encoders differ in architecture as well. We are explicit about this throughout. The headline comparison is therefore between encoder *families* as configured, and the controls of §4.5 are designed to separate the objective from the architecture and the wake supervision.

3.3 The matched-predictor fairness protocol

The comparison rests on a single principle: every baseline is the same three-stage pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields; an identical autoregressive transformer predictor rolls it forward one step at a time given $\mathbf{c}$; and an identical probe family maps the predicted latent to a physical observable for evaluation. The predictor architecture, its training recipe, the parametric conditioning, and the probe family are held fixed across the three encoder families. The latent dimension is matched within each family: JEPA at $d \in \{32, 64\}$, the reconstructive autoencoder at $d \in \{3, 32, 64\}$ to include both the published $d = 3$ recipe and matched-capacity variants, and POD at $d \in \{16, 32, 64\}$. This isolation is the only fair way to attribute forecast quality to the representation rather than to incidental differences in the predictor or its training.

3.4 Forward physical closure

The figure of merit is forward physical closure. Given a ground-truth $\mathbf{c}$ and an initial latent, the predictor is rolled recursively to $H = 16$, and at each step the probe maps the predicted latent to each observable. A small closure error means the encoder produces a latent whose dynamics, propagated by the shared predictor, retain the information the probe needs to recover the physical quantity; a large error means either that the encoder discarded that information or that the predictor lost it during rollout, a distinction the drift diagnostic below resolves. We report closure on the held-out test_b and test_c encounters as the mean absolute error against the simulation, with bootstrap 95% confidence intervals over encounters and the coefficient of determination R^2. The held-out forecast R^2 (Markov rollout at $H = 16$) is reported in Table 2, and the representational closure (probe on the simulation-encoded latent) in Table 2.

3.5 Two mechanistic diagnostics

Latent drift. A predictor with low one-step error need not produce a useful long rollout, because each prediction is fed back as the next input and small geometric errors accumulate. We quantify how far a rollout has left the training manifold by the Mahalanobis distance of the rolled-out latent at horizon H, measured against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has gone out of distribution, so that the probe is then queried outside its support.

Conditioning-only floor. Because the predictor is conditioned on $\mathbf{c} = (G, D, Y)$, a natural concern is that the forecast quality is carried by the conditioning rather than by the latent. We bound this with a kernel-ridge regressor that maps $\mathbf{c}$ directly to each observable at the impact frame, with no latent input, fit on training and evaluated on the held-out splits. This is the floor any learned encoder must clear on held-out cases to demonstrate that its latent contributes beyond what is already implicit in the parameters.

Table 1: Encoder families compared under the matched-predictor protocol of §3.3. All three share one autoregressive transformer predictor, training recipe, conditioning on $\mathbf{c} = (G, D, Y)$, and probe family; only the encoder differs. "Aux. heads" lists the observable supervision attached to the encoder. The architecture and auxiliary-head columns differ across families, which is why the headline result is attributed to the predictive *family* and isolated only by the controls of §4.5.

Family	Objective	Encoder	Latent dim. d	Aux. heads
JEPA (this work)	predictive (latent)	CNN + ViT	32, 64	lift + wake
Reconstructive AE	reconstructive (field)	CNN	3, 32, 64	lift
POD	linear projection	—	16, 32, 64	—

4 Results

4.1 Forward physical closure across encoder families

Table 1 summarises the three encoder families and the protocol. We distinguish two questions. Does the latent, encoded directly from a held-out field, carry each observable (representation quality, Table 2a)? And does the predictor, rolled out from the impact frame, forecast it (forecast quality, Table 2b)? The first isolates the encoder; the second tests the encoder and the learned dynamics together. The predictive latent leads on both, and the forecast is the load-bearing test. On representation, the wake separation is widest: at $d = 64$ the predictive wake-enstrophy error is 29.83 against 72.90 for the reconstructive autoencoder and 89.31 for the linear basis, a factor of 2.4 and 3.0, with the smallest positive and negative circulation errors. The gap narrows on the scalar forces, which all three families carry more evenly, although the predictive latent still leads on C_L (0.624 against 0.989 and 0.872). The integrated impulse I_y is where the linear basis is most competitive: on the in-distribution training fit (Table 6) POD attains the highest I_y score of any family, an expected consequence of its energy-optimal projection, although on held-out error the predictive latent still leads there. We flag this train-versus-held-out reversal because it is exactly why a held-out coefficient of determination, not the training fit, must head the comparison.

On the forecast (Table 2 and Figure 5), evaluated as the held-out R^2 at $H = 16$ of the Markov rollout, the predictive latent is the only representation whose wake-enstrophy forecast stays above the predict-the-mean floor: $R^2 = 0.449$ at $d = 64$ (0.214 at $d = 32$), against -0.478 for the reconstructive autoencoder and -0.089 for the linear basis, both worse than forecasting the held-out mean. It also leads the mean over the six observables (0.445 versus 0.147 for both the $d = 64$ reconstructive and linear bases) and the negative circulation. The training-set fit (Table 6, mean 0.835 at $d = 64$) measures in-distribution accuracy, not generalisation; the held-out forecast is markedly lower in absolute terms, as it must be, but it is where the families separate. On the OOD test_c set ($G = +4$) the predictive wake-enstrophy forecast remains positive ($R^2 = 0.33$ at $d = 64$, 0.45 at $d = 32$) while every other family falls to between -1.1 and -1.5; the force coefficient C_L is the only observable any family forecasts well there ($R^2 = 0.79$), consistent with the three-dimensional observability boundary of §2.1.

All held-out closure numbers carry 2000-resample bootstrap confidence intervals (reported in full in Appendix A). Taken one observable at a time these marginal intervals are wide, because they are dominated by the encounter-to-encounter difficulty of the held-out set that every model shares: the predictive forecast wake-enstrophy R^2 point estimate of 0.449 has interval $[-0.96, 0.79]$, and the forecast wake error 41.3 has interval $[29.1, 59.2]$ against the reconstructive 73.3 $[50.2, 98.4]$. Pairing the comparison per encounter removes that shared difficulty and is the appropriate test (Table 7): the predictive latent carries the smaller wake-enstrophy error on 31 of 42 held-out encounters under direct encoding (paired mean improvement 43.1, 95% CI $[23.5, 66.0]$, one-sided sign $p = 1.4 \times 10^{-3}$) and on 27 of 42 under forward forecast (paired mean

Table 2: Held-out closure across encoder families at $H = 16$ on the $n = 42$ test_b encounters, in two modes. *(a)* Representational closure: the mean absolute error when each family's per-frame probe is applied to the simulation-encoded latent (the latent encoded directly from the held-out field), smaller is better. *(b)* Forecast closure: the coefficient of determination R^2 of the Markov rollout from the impact frame against the simulation observable, larger is better (the variance baseline is the held-out split mean, so $R^2 < 0$ is worse than predicting that mean). Best per column in bold; the forecast panel gives the row mean at right. The wake-enstrophy and circulation columns are in native units, the force coefficients dimensionless. This held-out forecast R^2, not the training-set fit of Table 6, is the reported closure quality. The predictive latent leads in both modes: its representational wake-enstrophy error is 2.4 to 3.0 times lower than the reconstructive and linear families, and its forecast wake-enstrophy R^2 is the only one above the predict-the-mean floor; the paired per-encounter test is in Table 7 and full probe values with bootstrap CIs are in Appendix A.

(a) Representational closure: mean absolute error

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.
JEPA	64	**0.624**	0.301	**1.901**	**29.83**	**0.801**	**0.715**
JEPA	32	0.675	**0.246**	2.976	37.99	1.084	1.061
Fukami	3	1.074	0.384	2.327	51.70	1.110	1.886
Fukami	32	1.038	0.368	2.195	63.94	0.946	1.789
Fukami	64	0.989	0.340	2.421	72.90	1.057	1.904
POD	16	1.168	0.317	2.235	89.14	1.645	1.681
POD	32	0.937	0.269	2.608	81.82	1.650	1.617
POD	64	0.872	0.291	2.245	89.31	1.660	1.800

(b) Forecast closure: coefficient of determination R^2

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	mean
JEPA	64	0.723	0.634	0.056	**0.449**	0.201	**0.607**	**0.445**
JEPA	32	**0.751**	**0.648**	-0.082	0.214	0.087	0.552	0.362
Fukami	3	0.632	0.226	0.115	-0.082	0.258	-0.048	0.184
Fukami	32	0.641	0.239	**0.205**	0.007	**0.420**	0.161	0.279
Fukami	64	0.729	0.406	0.043	-0.478	0.182	0.003	0.147
POD	16	0.499	0.364	-0.515	-0.129	-0.230	0.340	0.055
POD	32	0.714	0.558	-0.256	0.042	-0.066	0.394	0.231
POD	64	0.683	0.539	-0.813	-0.089	0.057	0.507	0.147

improvement 32.0, 95% CI $[10.8, 54.8]$, sign $p = 4.4 \times 10^{-2}$), so the primary wake comparison is significant once the shared difficulty is cancelled. All six observables are reported in both modes, not a selected subset, so the consistency of the ordering across every observable is itself part of the evidence. The reconstructive *representational* wake-enstrophy R^2 interval is entirely negative ($[-4.62, -0.12]$), a cleaner statement that the reconstructive latent fails to carry wake structure on held-out cases, and the decisive confirmation is the mechanistic evidence of §4.3 (the drift ratio, the topological generator count at $p = 4 \times 10^{-8}$, the transport alignment, and the scale-resolved wake-enstrophy tracking), where the families separate unambiguously.

The conditioning-only floor (Table 4) confirms that the held-out result is not an artefact of the parametric conditioning the predictor shares. A kernel-ridge map from $\mathbf{c} = (G, D, Y)$ to the impact-frame observables interpolates the training points well, as any flexible kernel in a three-dimensional input will, but collapses on test_b and goes negative on test_c for five of six observables. The parameters alone cannot generalise outside the training envelope, so the held-out closure is doing representational work that the conditioning cannot replicate.

Table 3: Latent-drift diagnostic on test_b at $H = 16$. The Mahalanobis distance of the rolled-out latent is reported alongside that of the simulation-encoded latent in the same reference distribution; their ratio $r = d_{\text{Mahal, Markov}}/d_{\text{Mahal, DNS}}$ measures how far the autoregressive rollout has left the training manifold. A ratio near one means the rollout stays in distribution.

Baseline	d	$d_{\text{Mahal, ref}}$	$d_{\text{Mahal, rollout}}$	ratio r
Fukami	64	2.92	28.96	**9.90**
JEPA	64	8.08	6.84	0.85
JEPA	32	5.43	4.65	0.86
POD	64	7.74	6.27	0.81

Table 4: Conditioning-only floor. A kernel-ridge regressor maps the gust parameters $\mathbf{c} = (G, D, Y)$ directly to each observable at the impact frame, with no latent input; R^2 is reported on each split. This is the floor any learned encoder must clear on held-out cases to show its latent contributes beyond the parameters. The conditioning floor interpolates the training points but collapses on the held-out splits, going negative on test_c for five of six observables. The last column is JEPA's held-out forecast R^2 (Table 2, d=64, test_b), the like-for-like comparison. The predictive forecast clears the floor on the force coefficients and the circulations; on wake-enstrophy the H=16 forecast (0.449) is about level with the parameter floor (0.482), while JEPA's representational closure (probe on the encoded latent, $R^2 = 0.754$, Table 2) clears it comfortably.

Observable	n_{train}	train R^2	test_b R^2	test_c R^2	JEPA d=64 test_b R^2
C_L	226	0.895	0.303	0.547	0.723
C_D	226	0.941	0.386	-2.459	0.634
I_y	226	0.422	-0.301	-1.335	0.056
wake-enstrophy	226	0.806	0.482	-1.252	0.449
circ. pos.	226	0.792	-0.032	-0.249	0.201
circ. neg.	226	0.799	0.568	-4.262	0.607

4.2 Horizon dependence

A dynamics paper should not rest on a single horizon. Comparing recursive (Markov) rollout against a one-shot prediction at horizon H, the two agree to within the encounter-level scatter for $H \leq 16$ and diverge for $H \geq 32$ as the recursive error compounds, which places the reported $H = 16$ inside the regime where the rollout mode does not affect the conclusion and which matches the planning horizon relevant to the impact-instant load. The full decay of closure with horizon, swept over $H \in \{1, 4, 8, 16, 32, 64\}$ for every family on test_b and test_c (Figure 6), separates the families by the manner of their failure, not only its degree. The predictive latent degrades gracefully: its held-out wake-enstrophy forecast R^2 stays above 0.5 to $H = 16$ at both $d = 64$ and $d = 32$, then declines. The reconstructive and linear latents fail abruptly: their wake-enstrophy forecast R^2 is already below 0.5 at $H = 1$ and negative by $H = 16$, so they never hold a usable wake forecast. The force coefficient C_L, which all families carry, decays smoothly for all of them. The abrupt wake failure coincides with the rollout-drift onset of §4.3: the reconstructive latent leaves its manifold within a few steps, so its wake forecast collapses immediately rather than eroding.

4.3 The drift mechanism: distance, topology, and transport geometry

The closure gap is large enough that its mechanism deserves direct measurement.

Table 5: Objective × architecture controls (decisive experiment). Held-out wake-enstrophy closure and mean closure (R^2 at $H = 16$, test_b, Markov rollout, mean ± standard deviation over three encoder seeds) for the four cells crossing {predictive, reconstructive} with {CNN, CNN+ViT} at $d = 64$, with the auxiliary heads (current lift at $\delta = 0$ plus the 80-dimensional wake head) matched across all four cells, plus a predictive variant with the wake head removed. Predictive beats reconstructive on wake closure at both architectures (predictive versus reconstructive at CNN+ViT, and again at CNN), so the objective is not a confound of the architecture; removing the wake head from the predictive model collapses wake closure below the floor, so the wake supervision is also necessary. The CNN and CNN+ViT columns do not separate, so the ViT is not the driver.

Objective	Encoder	Aux. heads	wake-enst. R^2	mean R^2
predictive	CNN+ViT	lift + wake	$\mathbf{0.46 \pm 0.03}$	0.45
predictive	CNN	lift + wake	0.45 ± 0.06	**0.52**
reconstructive	CNN+ViT	lift + wake	0.16 ± 0.27	0.23
reconstructive	CNN	lift + wake	0.29 ± 0.05	0.42
predictive	CNN+ViT	lift only	-1.03 ± 0.29	0.09

Distance. The latent-drift diagnostic (Table 3) measures how far each rollout leaves its training manifold. After $H = 16$ steps the reconstructive rollout sits an order of magnitude further from its training distribution than its simulation-encoded counterpart (ratio 9.90), while the predictive and linear rollouts remain within it (ratios 0.85 and 0.81). This is the mechanism behind the closure gap: the probe attached to a drifted rollout is queried outside the support it was fit on, so its output degrades for reasons unrelated to the predictor's one-step quality. A reconstruction objective places latent codes wherever the decoder finds convenient and does not constrain the geometry to be predictable in time; the predictive objective, regularised against collapse, produces a latent that supports stable rollout by construction. The pre-impact inference of the impact lift makes the same point from the encoder side: an oracle probe on the simulation-encoded predictive latent recovers the impact C_L at $R^2 = 0.683$ at every lead time, whereas the same oracle on the reconstructive latent is negative (-0.185), a representation failure rather than a probe failure, since no probe can recover information the latent does not carry. Rolling the predictive latent forward through the predictor recovers about 0.60 of that skill from five frames before impact and about 0.35 from ten, several frames of usable forecast horizon beyond a direct wall-pressure read at the same lead times.

Topology. A single distance understates the geometric content of the result. A gust encounter is, in state space, a closed cycle: the flow departs the baseline limit cycle, executes the LEV-formation and shedding excursion, and returns [19]. We characterise this with persistent homology, which identifies the persistent one-dimensional cycle (H_1 generator) of a trajectory by a Vietoris–Rips filtration [19], applied here to four latent point clouds per encounter: the simulation-encoded predictive latent, its rollout, the simulation-encoded reconstructive latent, and its rollout (Figure 7). The coordinate-free signature is not the survival of a single generator's lifetime, which is confounded by scale (the reconstructive rollout drifts into a large diffuse cloud that still supports a long-lived but non-canonical loop, so the rollout-to-encoded lifetime ratio is near one for both families and does not separate them). It is the *generator count* of the simulation-encoded latent itself: the predictive encoding produces a clean single cycle (median one significant H_1 generator on test_b, 55% of encounters with exactly one), while the reconstructive encoding fragments into several (median 3.5, 71% with three or more; Mann–Whitney $p = 4.4 \times 10^{-8}$, and the same direction on test_c, $p = 0.04$). The predictive latent has the clean limit-cycle topology of the physical encounter; the reconstructive latent does not. This is consistent with the curvature geometry of §4.4: the impact frame is a curvature minimum of the predictive trajectory, a smooth pass-through rather than a sharp corner, so the cycle it traces is a single well-formed loop. The result at $d = 32$ is the same (single predictive loop versus reconstructive fragmentation).

Transport geometry. The geometric reason the predictive rollout stays on the manifold can be made physical. Euclidean distance between vorticity fields is insensitive to the spatial transport of structures, whereas an unbalanced optimal-transport (OT) distance measures the cost of rearranging one field into another and so tracks advection of the LEV and shear layer [23]. Computing the OT distance between simulation fields along an encounter gives a physical transport geometry against which the latent metric can be checked: the prediction is that distances in the predictive latent track the OT geodesic more faithfully than distances in the reconstructive latent, so that a single predictor step corresponds to a physically smooth, transport-consistent change and iterating it does not leave the data manifold. The prediction holds: the per-encounter Spearman correlation between the latent distance matrix and the simulation OT-distance matrix on test_b is 0.63 for the predictive latent at $d = 64$ (0.61 at $d = 32$) against 0.45 for the reconstructive latent, a margin of 0.18, with the predictive latent more faithful on 36 of 42 encounters. The predictive latent metric is the better isometry of the physical transport geometry, so a single predictor step is a transport-consistent move and iterating it stays on the manifold. We report this per encounter rather than pooled across encounters: pooling conflates the within-encounter geometry with a between-encounter latent-norm scale that the drift-prone reconstructive latent inflates, which reverses the pooled statistic and is itself a symptom of the same drift.

4.4 Where closure holds and fails: parameter and phase dependence

Which gust axes the latent resolves. Probing the predictive latent for the gust parameters on test_b recovers the gust strength G and the core diameter D ($R^2 = 0.46$ and 0.80) but barely resolves the wall-normal offset Y ($R^2 = 0.10$). This is a property of the data rather than of the architecture: the training mass concentrates near $Y \approx 0$, so encounters at different offsets within the envelope are weakly separated along Y, and the latent inherits that weakness. Any deployment-time parameter estimate should treat Y as a nuisance variable the encoder resolves only weakly.

Phase relative to the baseline cycle. Because the undisturbed wake is a limit cycle (§2.1), an encounter is naturally described by its phase and amplitude relative to that cycle, the same reduction used to design gust-mitigation control for these flows [6]. Reading the predictive latent this way makes “recovery” precise: it is the return of the latent trajectory to the baseline limit cycle. The predictive latent of the undisturbed case does trace a closed periodic orbit (Figure 8): concatenating the four no-gust episodes, the return distance is 1.0% of the orbit diameter, and the four episodes overlap the same loop, with a shedding period of about 56 frames (2.8 convective times, Strouhal number near 0.36). A phase advances monotonically along it (via the Hilbert analytic signal of the two leading latent principal components), though the orbit lives in more than two dimensions, so the phase is a reduction rather than a complete coordinate. A gust encounter departs this orbit and is carried back toward it under the predictive rollout while the reconstructive rollout departs further: the median return-to-orbit distance on test_b is smaller for the predictive latent at every horizon, and the separation is bootstrap-robust at $H = 64$ (predictive 8.70 versus reconstructive 9.96, 95% confidence interval on the paired difference $[1.13, 3.25]$), marginal at $H = 32$. One caveat is load-bearing and we state it: the simulation gust trajectories do not themselves fully return to the baseline orbit within the 120-frame window, so the comparison measures the relative drift direction (the predictive rollout contracts toward the orbit, the reconstructive one expands away), not a completed physical recovery. The result is the same at $d = 32$. This limit cycle and the persistent cycle of the previous subsection are the same object viewed dynamically and topologically.

Where the error concentrates. The first-order answer to where closure holds is already in the per-axis probe above: the predictive latent resolves G and D but weakly resolves Y, so its residual error is expected to concentrate at off-midplane offsets. A full per-stratum closure table, broken out by gust regime and encounter phase as in the source characterisation of these flows [7], is left to a study with more cases per cell than the present split affords; the two-tier interior and boundary test_b reporting already exposes the corner-case behaviour that such a table would isolate.

4.5 Isolating the cause: objective, architecture, and auxiliary-head controls

The results so far establish that the predictive latent *family*, as configured, closes the forward observables better than the reconstructive and linear families, and that the mechanism is reduced rollout drift. They do not yet establish that the *predictive objective in isolation* is responsible, because the predictive encoder differs from the reconstructive baseline in two further ways: its CNN+ViT architecture, and its 80-dimensional wake supervision. The latter is the more pointed confound, since the largest part of the reported advantage is on the wake observables that the predictive encoder alone was supervised on. Isolating the objective requires the controls in Table 5: a 2×2 crossing of objective {predictive, reconstructive} with architecture {CNN, CNN+ViT} at matched $d = 64$, with the auxiliary heads matched across all four cells, evaluated under the same predictor and probes; together with a predictive variant from which the wake head is removed. The decisive question is whether the predictive objective still closes the wake observables better than the reconstructive one once the architecture and the wake supervision are held in common.

The controls (Table 5, three encoder seeds per cell) give a two-part answer, and we state both. First, the predictive objective improves wake closure at *both* architectures with the auxiliary heads matched: the held-out wake-enstrophy R^2 is 0.46 ± 0.03 for the predictive CNN+ViT against 0.16 ± 0.27 for the reconstructive CNN+ViT, and 0.45 ± 0.06 for the predictive CNN against 0.29 ± 0.05 for the reconstructive CNN. The objective, not the architecture, carries the wake advantage: the CNN and CNN+ViT columns do not separate (the predictive CNN even leads on the mean-over-observables R^2, 0.52 against 0.45), so the ViT is not the driver. Second, and equally important, the wake-observable supervision is necessary: the predictive model with the wake head removed collapses to a wake-enstrophy R^2 of -1.03 ± 0.29, far below the floor and below every reconstructive cell. The wake-closure result therefore requires the predictive objective *and* the wake supervision together; it is not a property of the objective in isolation, nor of the architecture, nor of the wake head attached to an arbitrary objective (the reconstructive cells carry the same wake head and do not reach the predictive closure). We claim, accordingly, that the predictive objective improves forward wake closure over the reconstructive objective at matched architecture and matched supervision, while being explicit that the auxiliary wake head is a necessary component of the configuration that achieves it. The large seed variance of the reconstructive CNN+ViT cell (± 0.27) is itself consistent with the drift picture: without a forward-predictable latent geometry, wake closure is unstable across initialisations.

4.6 Physical-space interpretation

Forward closure is a statement about numbers; the encounter is a statement about vortices. Decoded and reconstructed vorticity fields on held-out encounters (Figure 11) connect the two, with one methodological caution and one new metric.

The caution is that field-space similarity scores are misleading here. On a held-out interpolation encounter the reconstructive autoencoder attains a high structural similarity yet a peak reconstructed vorticity magnitude of only 0.36 against a target peak near 9.7: it has collapsed to a near-uniform mean field and lost the vortex-impingement signature, and the high similarity merely reflects bulk agreement with a field that is itself mostly quiescent away from the impact region. The predictive latent localises the impingement at the right position and sign (peak magnitude 5.31) but its visualisation decoder is blurry by construction, and the linear basis is amplitude-accurate (peak 4.10) since its objective is amplitude-optimal.

The new metric resolves the caution. An unbalanced optimal-transport distance between the reconstructed and the simulation fields, computed for the positive and negative vorticity separately and summed [23] (Sinkhorn divergence, fields pooled to 48×24 for tractability), penalises the collapsed reconstruction correctly and reorders the comparison where structural similarity misleads (Figure 9). At the impact frame on test_b the transport distance is largest for the reconstructive autoencoder (11.25) and smaller and comparable for the predictive decode (9.90) and the linear basis (9.95). The instructive case is the linear basis: it carries the *highest* structural similarity of the three (0.69 against 0.65 for the predictive decode), which would wrongly crown the energy-optimal reconstruction as the best, yet under transport it is no better than the blurry predictive decode and far better than the collapsed autoencoder, which is the physically correct ordering. We therefore lead with the transport distance and report similarity alongside for continuity. One honest boundary: on the OOD test_c set ($G = +4$) the transport ordering breaks (the reconstructive distance falls below the predictive one), so the transport advantage is an in-envelope result, of a piece with the three-dimensional observability boundary that degrades every metric there.

Finally, the wake observables tie to the structures that carry the load. A Gaussian scale decomposition (large scale at filter width $\sigma/c = 0.05$, following [18, 17]) separates the large-scale vortical structures responsible for the lift peaks from the fine scale, applied to the simulation field and to each family's reconstruction (Figure 10). On test_b the large-scale wake enstrophy of the predictive reconstruction tracks the simulation closely (Pearson correlation 0.89 at impact and 0.91 at $H = 16$, relative error 0.22), while the reconstructive autoencoder collapses it (correlation 0.23 and 0.61, relative error near 0.8, retaining only 16 to 20% of the large-scale amplitude and almost none of the small-scale energy): it smooths the leading-edge vortex and shear layer away. The linear basis has comparable large-scale correlation to the predictive latent but worse amplitude (relative error 0.34 to 0.38), so the claim is specific: the predictive latent tracks the simulation's large-scale wake structure better than the reconstructive autoencoder on both correlation and amplitude, and better than the linear basis on amplitude. Across the staged encounter the simulation large-scale enstrophy rises from the gust-induced leading-edge flux through the LEV roll-up and peak-load shedding; the predictive reconstruction follows that rise while the reconstructive one stays flat. This is where the wake-enstrophy advantage of §4.1 becomes a statement about the leading-edge vortex and shear layer rather than about a scalar. The result holds at $d = 32$; on the OOD test_c set the predictive tracking degrades (0.91 to 0.65 from impact to $H = 16$), as expected where the flow becomes three-dimensional and the mid-plane decomposition is itself incomplete (§2.1).

5 Discussion

5.1 What the predictive latent retains, and why

The central finding is about dynamic closure, not reconstruction. Under a shared predictor, parametric conditioning, and probe family, the predictive latent family keeps the forward evolution of six physical observables closer to the simulation than the reconstructive and linear families do, and the advantage is concentrated on the spatially distributed wake observables that integrate over the shedding region. The mechanism is geometric. A reconstruction objective fixes the latent only up to a diffeomorphism [23, 12], so the decoder is free to place codes anywhere it finds convenient and nothing constrains the latent geometry to be predictable in time; under autoregressive rollout the reconstructive latent therefore leaves its own training manifold, and a probe queried on that drifted state fails for reasons unrelated to the predictor's one-step accuracy. The predictive objective, regularised against collapse, instead organises the latent so that the encounter is a closed cycle that survives rollout and so that latent distance tracks the physical transport geometry of the flow. Iterating the predictor therefore does not leave the manifold, the property that data-driven state-space and Koopman models of coherent-state dynamics on invariant manifolds rely on [5]. The manifold departure of the reconstructive rollout is the parametric-gust counterpart of the catastrophic-divergence tail reported for reconstruction-based latent dynamics in controlled wakes [21]; the predictive objective is the common cure. The wake observables are the sharpest discriminator precisely because they can change while the scalar forces do not: a representation can carry the lift signature of an encounter without carrying the vorticity redistribution that produces it, and the reconstructive latent does exactly that, recovering the impact lift poorly from its own oracle while the predictive latent recovers it well.

This organisation survives a halving of the latent dimension, which is the basis for reading it as a low-dimensional predictive state rather than an artefact of capacity. At $d = 32$ the predictive latent matches the $d = 64$ latent on representation quality, the held-out wake-enstrophy representation R^2 is 0.74 against 0.75, and on every mechanism diagnostic: it traces the same single persistent cycle rather than fragmenting (§4.3), its latent metric aligns with the physical transport geometry to within the $d = 64$ margin (rank correlation 0.61 against 0.63), it returns to the baseline limit cycle under rollout (§4.4), it retains the same large-scale wake structure (§4.6), and its rollout-drift ratio is 0.86 against 0.85 (Table 3). The one quantity that degrades is the in-distribution wake forecast, where the $H = 16$ closure R^2 falls from 0.45 to 0.21 (Table 2): halving the dimension costs forecast sharpness on the wake observables while leaving the geometric mechanism intact. On the out-of-distribution test_c set the $d = 32$ latent is in fact marginally the more robust of the two, 0.45 against 0.33 on wake enstrophy. The physics the predictive objective organises into the latent, the closed encounter cycle and its transport-consistent metric, is therefore a low-dimensional object already present at $d = 32$, and the additional dimensions at $d = 64$ buy in-distribution forecast precision rather than a qualitatively different representation. The parameter content of the two latents is consistent with this reading: the dominant gust-strength axis is encoded as well at $d = 32$ as at $d = 64$, while the core diameter and the wall-normal offset, the axes the data resolve least, are the ones whose recovery degrades when the dimension is halved, the same axes whose forward closure the extra dimensions sharpen.

5.2 What the comparison does and does not establish

We are deliberate about the scope of the claim. The predictive encoder differs from the reconstructive baseline in three respects at once: the training objective, the CNN+ViT architecture, and the wake supervision. The conditioning-only floor removes the shared parametric conditioning as an explanation, and the matched-architecture, matched-auxiliary 2×2 controls of §4.5 separate the remaining two. They give a two-part answer that we state in full. The predictive objective does improve forward wake closure over the reconstructive objective at both a CNN and a CNN+ViT encoder with the auxiliary heads matched, and the two architecture columns do not separate, so the gain is the objective and not the architecture. The wake supervision, however, is a necessary part of the configuration that achieves it: a predictive model with the wake head removed collapses below the predict-the-mean floor on wake closure. The honest claim is therefore that the predictive objective improves forward closure *when trained with wake supervision*, not that the objective in isolation does, and not that the wake head attached to an arbitrary objective does, since the reconstructive cells carry the same head and do not reach the predictive closure. The linear basis, for its part, remains genuinely useful: it is the most competitive family on the integrated impulse and is amplitude-accurate in reconstruction, so the result is not that POD is obsolete but that it does not produce a latent whose rollout closes the wake observables.

5.3 Limitations

Three limitations bound the claims. First, the wall-normal offset Y is weakly resolved because the training mass concentrates near $Y \approx 0$; this is a data limitation, and Y-conditional statements should be read with it in mind. Second, the $|G| = 4$ extrapolation degrades for a physical reason, not merely a statistical one: the encoder observes a single mid-plane vorticity slice, and at $|G| = 4$ the true dynamics become three-dimensional [7], so test_c is best read as the observability boundary of a mid-plane representation rather than as a parametric-generalisation claim. Third, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss; a reader expecting sharp reconstructions from a predictive latent will be disappointed, which is the explicit trade the objective makes. To these we add, as a fourth, the co-necessity of the wake supervision established in §5.2: the predictive objective improves forward closure in the configuration tested, with the wake head as a necessary component, not as a property of the objective stripped of all auxiliary supervision.

5.4 Pathway to model-based control, and a pilot that does not yet close

The predictor is, by construction, the latent dynamics function a model-based controller plans against, and three properties of the architecture support that use: closure holds out to the planning horizon relevant to the impact-instant load, the latent is two to three orders of magnitude cheaper to evolve than the full field, and the rollout stays on the training manifold, which is the property that determines whether long-horizon plans remain meaningful since a planner queries the predictor on states reached by its own rollout. Because the encoder is unconditional, the latent itself is the observable a controller has access to, and the gust parameters enter only the predictor; replacing the parametric conditioning with an actuation channel is a change of input rather than of architecture, and the phase-amplitude reading of §4.4 connects the predictor directly to the sensitivity-function control designed for these flows [6]. In the action-conditioned world-model reading of §1, the present intervention is the gust; an actuation channel enters at the same point, the predictor conditioning, and the same forward-closure machinery applies with the control as the action rather than the disturbance. The predictor is the world model a controller would use; it is not itself a controller, and the pilot below shows the gap.

We also report, as a limitation rather than a result, a closed-loop pilot that chains a wall-pressure state estimator, the predictor rollout, and an observable probe. A temporal-convolutional estimator recovers the latent from a pre-impact pressure window well (held-out $R^2 \approx 0.87$ at eight sensors), but the assembled loop does not meet the tolerances set in advance: at the design horizon only a small fraction of held-out encounters fall within the C_L and I_y tolerance bands, and none of the pre-registered gates are met. The bottleneck is the rollout rather than the estimator, since the within- tolerance fraction is essentially unchanged when the oracle latent is substituted for the estimated one. The honest reading is that the representation and the pressure observability are in place but the rollout needs reinforcement, through more frequent re-estimation from pressure during the rollout or a longer scheduled-sampling horizon, before a usable controller follows. The estimator, sensor-placement, and closed-loop details are deferred to Appendix B.

5.5 Outlook

The contribution is narrow by design: a controlled comparison on a single parametric flow, anchored to forward physical closure rather than to in-distribution reconstruction, with a drift mechanism that explains the gap and a set of controls that bound its interpretation. The immediate next step is to run those controls and the horizon, topology, transport, and phase-amplitude analyses set out above, which together would either promote the claim to the predictive objective in isolation or sharpen it to the predictive family. Beyond that, applying the same matched-predictor closure protocol to other parametric flow datasets would test how general the predictive-versus-reconstructive trade reported here actually is.

6 Conclusions

We compared a predictive latent (JEPA), a reconstructive observable-augmented autoencoder, and a proper orthogonal decomposition basis as reduced states for parametric vortex-gust airfoil interactions at $\mathrm{Re} = 5000$, under a protocol in which a single autoregressive predictor, conditioning, and probe family were held fixed so that only the encoder varied, and judged each by forward physical closure rather than by reconstruction. On held-out encounters the predictive latent kept the wake observables closest to the simulation, by a factor of two to three over the other families on wake enstrophy, while the linear basis remained most competitive on the integrated impulse. The mechanism is rollout drift: the reconstructive latent leaves its training manifold by an order of magnitude over the forecast horizon, so that the wake information hidden from a reconstruction objective becomes recoverable from a predictive one, whereas the predictive and linear latents stay in distribution. Because the predictive encoder also differs in architecture and auxiliary supervision, we attribute the result to the predictive latent family and set out the matched controls that would isolate the objective. The predictor is the latent dynamics function of model-based control; a closed-loop pilot does not yet meet its tolerances, with the rollout rather than the state estimator as the bottleneck, and is reported as the boundary of what the present representation supports.

A Architecture, anti-collapse regularisation, and uncertainty quantification

Architecture detail. The encoder is a hybrid convolutional stem followed by a small vision transformer. Three downsampling convolutional stages map the single-channel 192×96 vorticity field to a 24×12 feature map at 256 channels, that is 288 spatial tokens, which a six-layer pre-norm transformer of width 256 with eight heads processes; the latent is read from a prepended class token through a one-layer projection whose output passes a BatchNorm, the layer the characteristic-function regulariser below requires. The encoder is unconditional and carries no information about the gust parameters. The predictor is a six-layer autoregressive transformer of width 384 with sixteen heads and dropout 0.1, with a causal mask and rotary temporal positions. The gust parameters $\mathbf{c} = (G, D, Y)$ and the within-encounter phase enter only the predictor, through adaptive layer normalisation initialised to the identity (AdaLN-Zero), so that at initialisation the predictor is the unconditioned dynamics and the conditioning is learned as a modulation. Training minimises the one-step latent prediction loss plus a scheduled-sampling rollout term and the regulariser below, with AdamW and bf16 mixed precision.

Anti-collapse regularisation. Because the encoder and predictor are trained end to end on latent-space prediction with no decoder, the latent is held away from a collapsed solution by an explicit regulariser rather than by a reconstruction term. We use SIGReg, the characteristic-function (Epps–Pulley) regulariser of the from-pixels joint-embedding line [14, 3], with M random

Table 6: Training-set fit, for reference only. Coefficient of determination R^2 for predicting the six observables from the rolled-out latent ($H = 16$) evaluated *on the training distribution*. This quantifies in-distribution fit and is *not* the headline; held-out closure is in Table 2. Best per column in bold; row mean at right.

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	mean
JEPA	64	**0.864**	**0.802**	0.562	**0.934**	**0.922**	**0.927**	**0.835**
JEPA	32	0.863	0.795	0.490	0.898	0.901	0.897	0.808
Fukami	3	0.553	0.408	0.191	0.079	0.130	0.033	0.232
Fukami	32	0.695	0.596	0.278	0.333	0.369	0.313	0.430
Fukami	64	0.671	0.589	0.335	0.277	0.336	0.352	0.427
POD	16	0.763	0.720	0.514	0.574	0.582	0.527	0.613
POD	32	0.763	0.748	0.690	0.589	0.618	0.621	0.671
POD	64	0.641	0.656	**0.799**	0.373	0.388	0.506	0.561

one-dimensional projections of the latent and a small weight (0.01) in the loss. As a safeguard the participation ratio of the latent is monitored every thousand iterations, with an automatic fall-back to a variance-covariance objective [4] if it indicates collapse while the parameter probe is still uninformative; on the runs reported here the participation ratio stays healthy and the fall-back does not trigger.

Uncertainty quantification. Every held-out coefficient of determination and mean absolute error in the paper carries three independent uncertainty signals, following the reporting protocol fixed with the split. The first is a bootstrap confidence interval from 2000 resamples over the held-out test encounters, which is the dominant source of scatter at the present sample size ($n = 42$ for test_b, $n = 24$ for test_c) and is wide for the individual wake observables, as noted in §4.1. The second is the variance across three encoder retrains from independent random seeds, reported as the standard deviation in the objective-times-architecture controls of §4.5. The third is a five-fold cross-validation over cases on the read-out (probe) step, which guards against a probe that overfits the particular held-out split. Together these approximate what a full five-fold case-level cross-validation of the entire pipeline would buy, at a fraction of the training cost; a single split with bootstrap intervals is the community standard for this class of study.

Training-set fit. For reference, Table 6 reports the in-distribution training fit that the held-out closure of §4.1 must be read against. The predictive latent attains the highest training fit (mean $R^2 = 0.835$ at $d = 64$), but the ordering of the families is established by the held-out closure of Table 2, not by this in-distribution number.

Paired closure across all observables. Table 7 reports the per-encounter paired comparison of §4.1 for all six observables in both modes, the source of the wake-enstrophy significance cited in the main text. Pairing cancels the encounter-to-encounter difficulty that widens the marginal intervals of Table 2.

B Sparse wall-pressure state estimation

At deployment the vorticity field is not available; only the wall pressure is. We therefore ask how much of the latent state, the gust parameters, and the impact lift can be recovered from a sparse set of wall-pressure taps, and whether routing the estimate through the predictive representation helps. The input is the span-averaged wall pressure on 192 surface taps, sub-sampled to K

Table 7: Paired held-out closure, predictive ($d = 64$) against reconstructive ($d = 64$) on the $n = 42$ test_b encounters at $H = 16$, for the representational mode (probe on the simulation-encoded latent) and the forecast mode (Markov rollout from impact). Δerr is the per-encounter mean of (reconstructive − predictive) absolute error in each observable's own units, so a positive value favours the predictive latent; the 95% interval is a 2000-resample paired bootstrap, and the sign test counts the encounters on which the predictive error is the smaller of the pair. Wake enstrophy separates the families in both modes by both the bootstrap interval and the sign test; all six observables are shown, not a selected subset.

Mode	Observable	Δerr	95% CI	sign test
repr.	C_L	+0.364	$[+0.070, +0.643]$	25/42, $p = 0.14$
repr.	C_D	+0.039	$[-0.052, +0.131]$	20/42, $p = 0.68$
repr.	I_y	+0.52	$[-0.028, +1.08]$	25/42, $p = 0.14$
repr.	wake-enst.	+43.1	$[+23.5, +66.0]$	31/42, $p = 1.4\times10^{-3}$
repr.	circ. pos.	+0.256	$[-0.117, +0.673]$	20/42, $p = 0.68$
repr.	circ. neg.	+1.19	$[+0.567, +1.93]$	30/42, $p = 4.0\times10^{-3}$
forecast	C_L	−0.021	$[-0.206, +0.152]$	20/42, $p = 0.68$
forecast	C_D	+0.066	$[+0.011, +0.128]$	26/42, $p = 0.082$
forecast	I_y	+0.264	$[-0.164, +0.666]$	26/42, $p = 0.082$
forecast	wake-enst.	+32.0	$[+10.8, +54.8]$	27/42, $p = 4.4\times10^{-2}$
forecast	circ. pos.	−0.018	$[-0.319, +0.297]$	19/42, $p = 0.78$
forecast	circ. neg.	+0.555	$[+0.067, +1.11]$	18/42, $p = 0.86$

sensors over a pre-impact window. This is a feasibility study, not a closed controller, and we report where it works and where it does not.

The map is nonlinear, and sensor placement is a region question. A linear ridge from all 192 taps to the latent recovers essentially nothing (held-out $R^2 \approx 0.03$); a temporal-convolutional estimator on a handful of taps reaches $R^2 \approx 0.88$, so the pressure-to-state map is strongly nonlinear and a linear observability analysis understates it. For sparse sensor placement [15] a target-conditioned greedy criterion matches the full 192-tap recovery with $K = 8$ to 16 taps and beats random placement, but the strongly collinear surface pressures do not admit a unique optimal tap set: the defensible output is a set of chord regions, clustered near the leading edge and the early suction surface, rather than specific taps.

In-distribution state recovery is strong, and the predictive latent is the most recoverable. With nonlinear estimators (temporal-convolutional, regularised multilayer perceptron, kernel ridge) the latent is recovered well on the in-distribution test_b set: the mean latent R^2 reaches 0.92 at $K = 8$ for the regularised perceptron, and 0.79 already at $K = 2$. Across encoder families at $K = 8$ (kernel ridge, Figure 12) the predictive latent is markedly the most pressure-recoverable: $R^2 = 0.87$ for JEPA at $d = 64$ and 0.84 at $d = 32$, against 0.79 for the reconstructive autoencoder at $d = 3$, 0.69 at $d = 64$, and only 0.43 and 0.16 for the POD coefficients at $d = 16$ and $d = 64$. The energy-optimal POD coefficients are the hardest to read from pressure, which is the deployment-side counterpart of the representation result in the main text. The gust parameters are also recoverable from pressure directly: a temporal-convolutional estimator reaches (G, D, Y) test_b $R^2 = (0.96, 0.95, 0.85)$ at $K = 16$, with G and D recovered even at $K = 2$ to 4 and the wall-normal offset Y the weakest, in keeping with its weak resolution throughout the paper. Recovering the parameters *through* a learned latent instead favours POD (G test_b $R^2 = 0.78$ against 0.46 for JEPA at $d = 64$), because the predictive latent is organised as a state, not as a parameter code, so its energy-aligned POD counterpart is the better intermediary for parameter read-out. The decisive caveat is generalisation: on the out-of-distribution test_c set ($G = +4$) every estimator and every family gives negative latent R^2 (between -0.5 and -2.8). Sparse pressure observability, like the rest of the model, holds inside the training envelope and fails at the three-dimensional observability boundary of §5.3.

Routing the lift estimate through the representation buys lead time. For the operational question of estimating the impact lift from pressure ahead of time, we compare a direct pressure-to-C_L regressor against the representation-mediated chain (pressure to the estimated impact latent to the C_L probe), as a function of the lead time before impact. At short lead, two frames before impact, the direct read is already accurate ($R^2 = 0.69$) and the representation adds nothing, because the imminent lift is already imprinted on the pressure. At long lead the direct read degrades below zero ($R^2 = -0.08$ at thirty frames), while the representation-mediated estimate holds up better, and the gap is sharpest out of distribution: thirty frames is too early for either, but at ten frames on test_c the representation-mediated estimate reaches $R^2 = 0.52$ against 0.24 for the direct read. The crossover is near five frames of lead: inside it, read C_L straight off the pressure; beyond it, route through the representation. The ceiling of the mediated chain is the $z \to C_L$ probe itself ($R^2 \approx 0.68$).

The closed loop is rollout-limited, not estimator-limited. Chaining the pressure estimator, the predictor rollout, and the observable probe into a closed loop does not yet meet a tight control tolerance: at $H = 16$ only 18% of test_b encounters fall within 10% of the true C_L, against a target of 80%. The informative point is that this gate fails even for an oracle that is handed the true impact latent and the true parameters (18% as well), so the tolerance is bounded by the irreducible error of the predictor and probe, not by the pressure estimator. The pressure chain matches the oracle to within a factor of 0.7 to 1.3 in physical-metric error at $K = 8$, so the estimated initial condition is not the bottleneck. The honest reading is that the representation and the pressure observability are in place in distribution, and the closed loop is limited by the rollout horizon rather than by the estimator; more frequent re-estimation from pressure during the rollout, or a longer scheduled-sampling horizon, is the route to a usable controller. We report this as a pathway, not a result.

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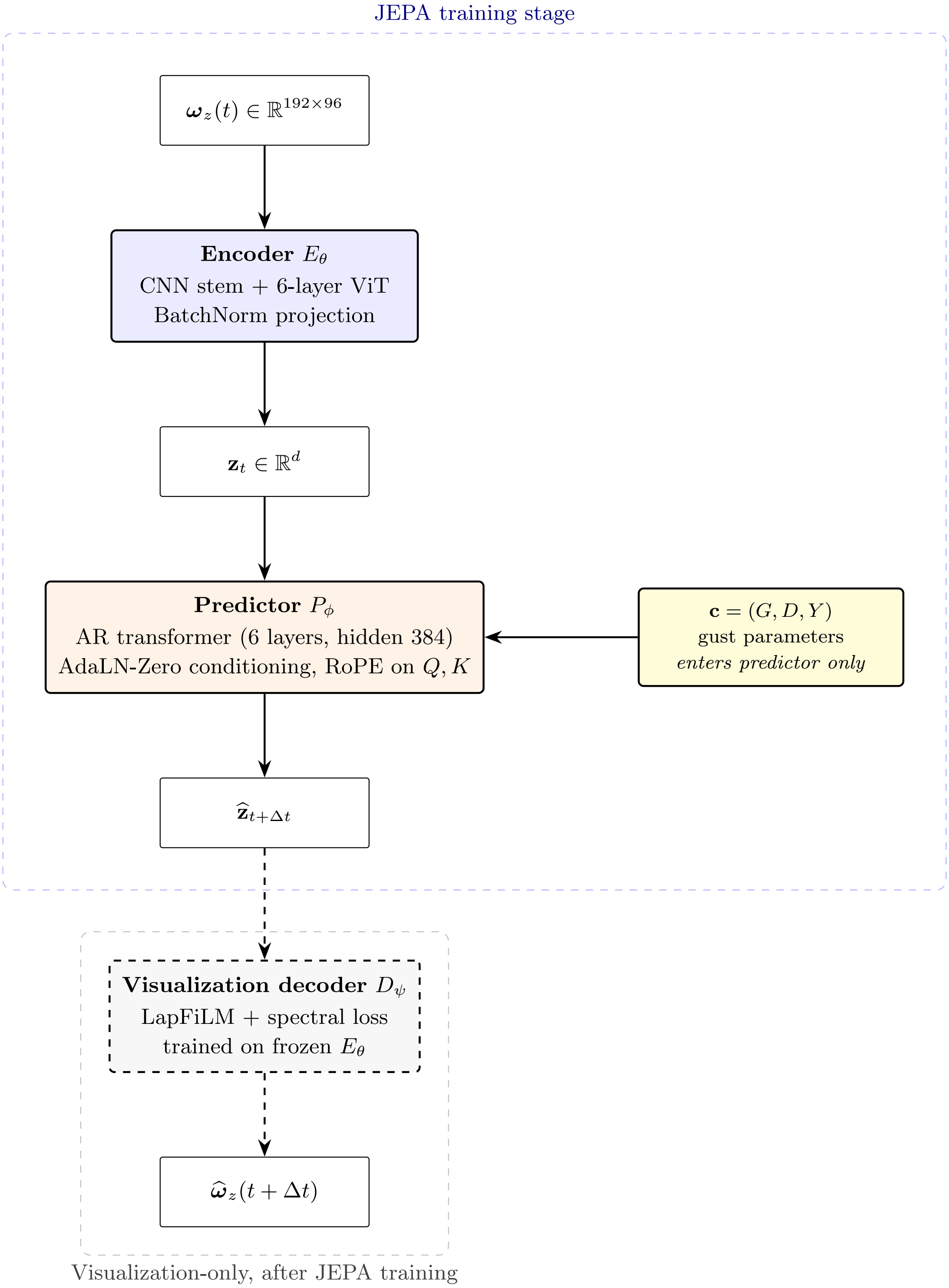

Figure 2: Predictive encoder and predictor. The encoder is unconditional and maps the mid-plane vorticity field to a latent $\mathbf{z}_t$. The autoregressive transformer predictor advances the latent given the gust conditioning $\mathbf{c} = (G, D, Y)$. The two auxiliary observable heads (lift and wake) and the visualisation decoder are attached to the frozen encoder and are never part of the latent-prediction loss.

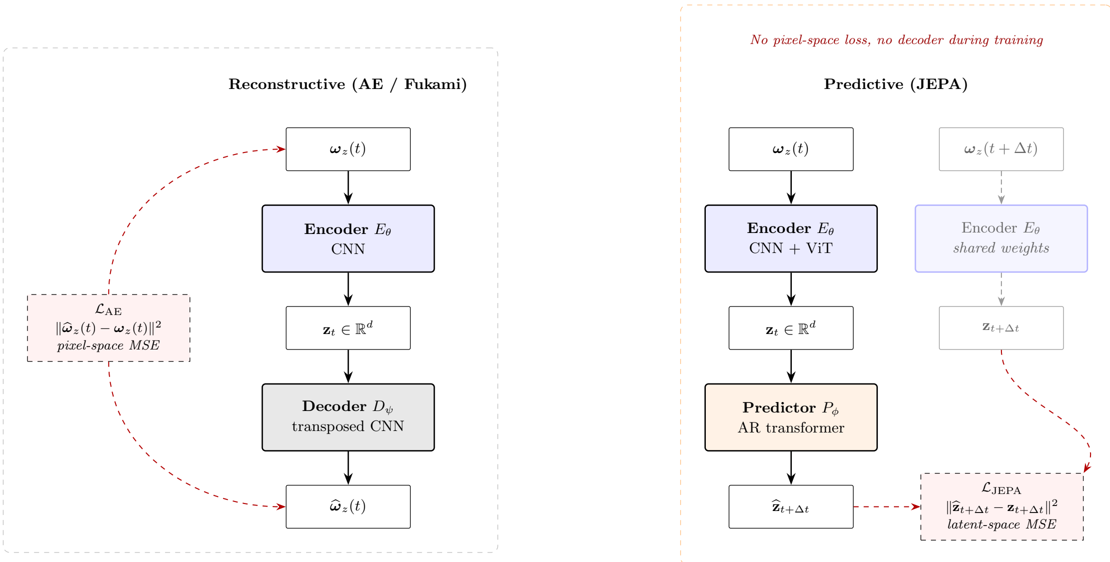

Figure 3: Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t, decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training.

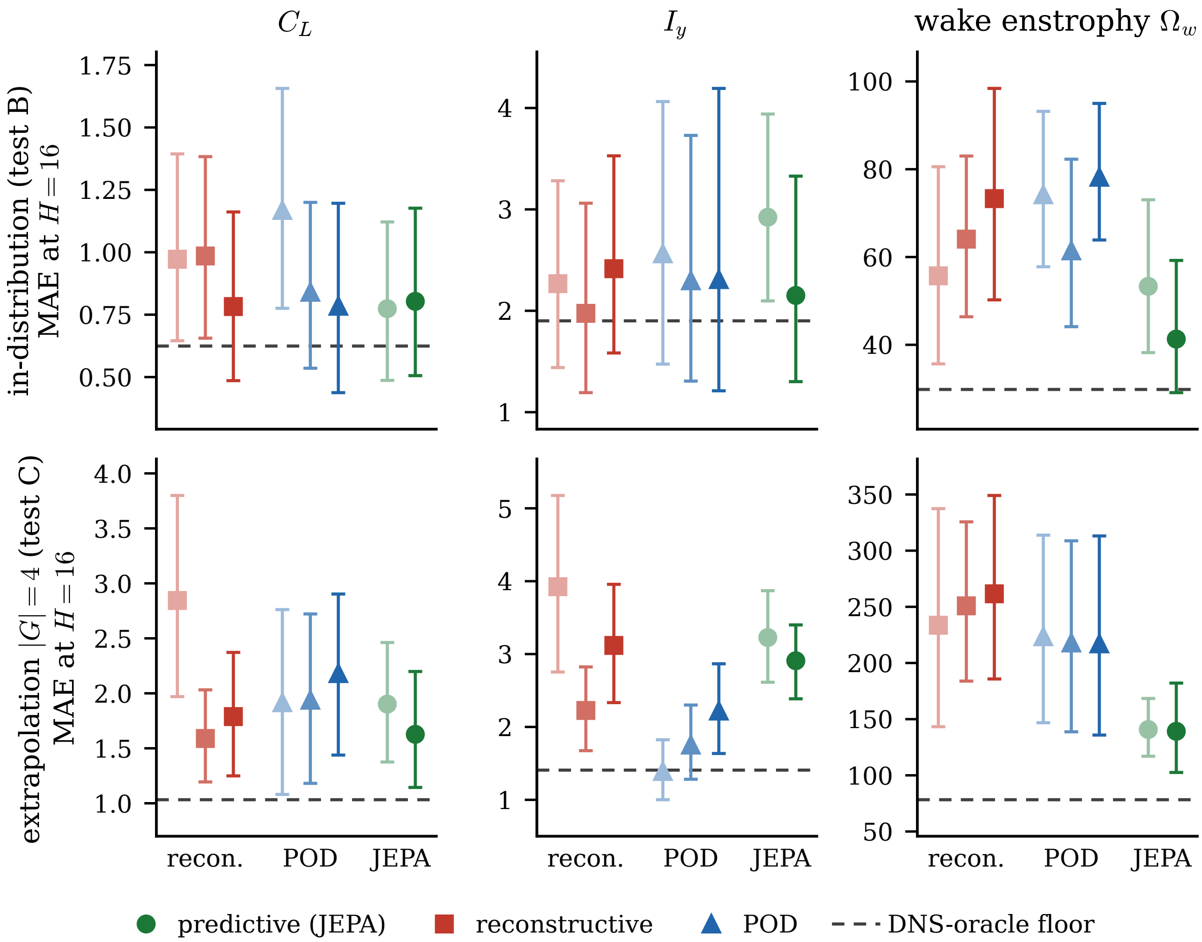

Figure 4: Held-out forward closure at $H = 16$ after gust impact, as the mean absolute error of the Markov rollout against the simulation, for the lift coefficient C_L, the wall-normal impulse I_y, and the wake enstrophy Ω_w (columns) on the in-distribution test set (test_b, top) and the $|G| = 4$ extrapolation set (test_c, bottom). Points are family means with bootstrap 95% confidence whiskers over $n = 42$ and $n = 24$ encounters, shaded from light to dark with the latent dimension within each family; the dashed line marks the DNS-oracle floor, the best closure achievable from the simulation-encoded latent without rollout. The wake panel shows the clearest separation: the predictive latent (JEPA) approaches the oracle floor while the reconstructive and linear families do not.

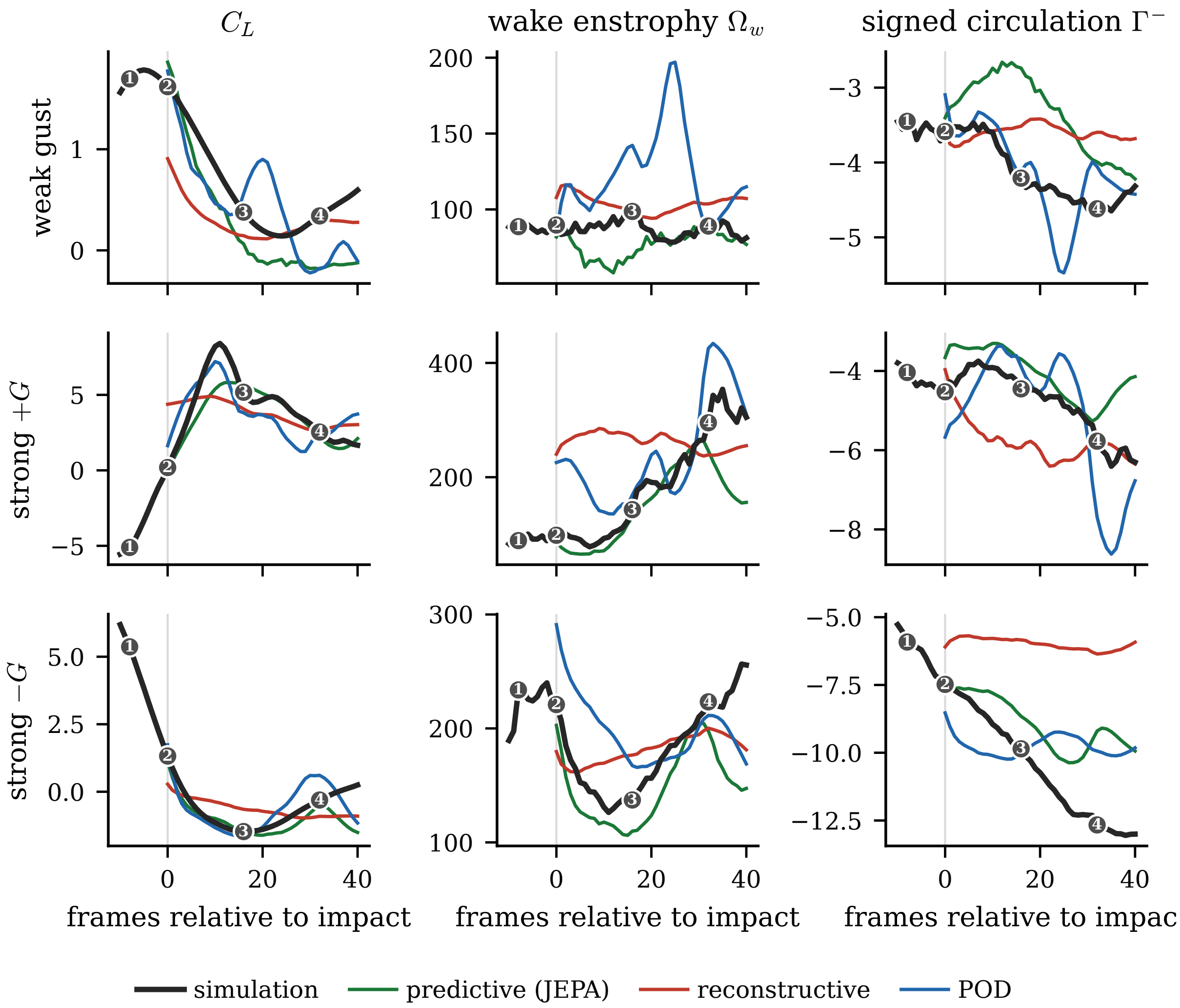

Figure 5: Predicted observables through three representative held-out encounters: a weak gust $(G, D, Y) = (-0.5, 1.0, -0.4)$, a strong positive gust $(+3.0, 1.0, +0.1)$, and a strong negative gust $(-3.0, 1.5, -0.1)$ (rows), for the lift coefficient, the wake enstrophy, and the negative circulation (columns). The simulation is the bold reference; the Markov rollouts of the predictive (JEPA), reconstructive, and POD latents are the coloured lines, started at the impact frame. The predictive rollout tracks the wake observables through the leading-edge-vortex peak and into recovery while the reconstructive rollout flattens and the linear rollout diverges. Numbered glyphs mark the four encounter stages used throughout: baseline, impact, peak load, and recovery.

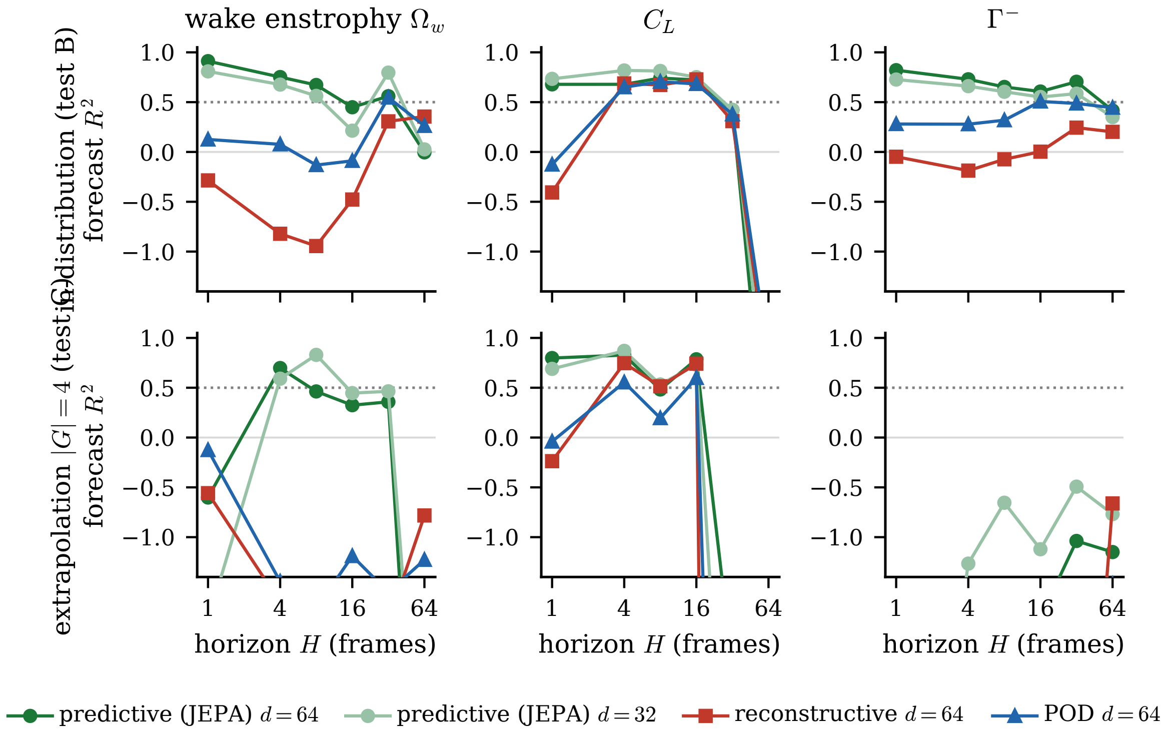

Figure 6: Held-out forecast closure (R^2 of the Markov rollout) versus horizon H for the wake-enstrophy, C_L, and negative-circulation observables, on test_b (top) and test_c (bottom), for the predictive latent at $d = 64$ and $d = 32$, the reconstructive autoencoder, and POD. The dotted line marks $R^2 = 0.5$. The predictive latent holds wake-enstrophy above the floor to $H = 16$; the reconstructive and linear latents are already below it at $H = 1$.

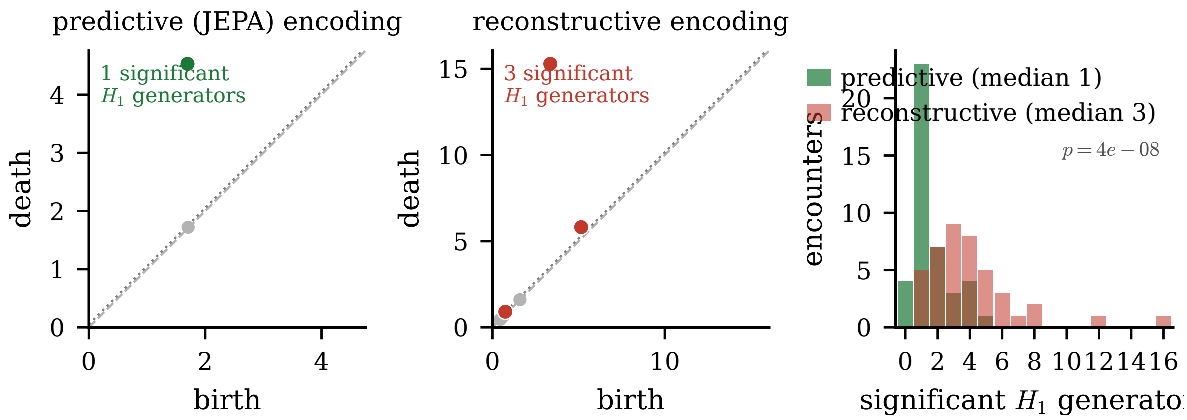

Figure 7: Persistent homology of the latent encounter on test_b. Left and centre: representative Vietoris–Rips H_1 persistence diagrams of the simulation-encoded predictive and reconstructive latents; coloured points are significant generators (lifetime above the dotted noise floor) and grey points are noise. Right: the distribution of the significant-H_1 generator count over the 42 held-out encounters, the scale-free signal. The predictive encoding is a single clean loop (median one generator) while the reconstructive encoding fragments (median three to four; Mann–Whitney one-sided $p = 4.4 \times 10^{-8}$).

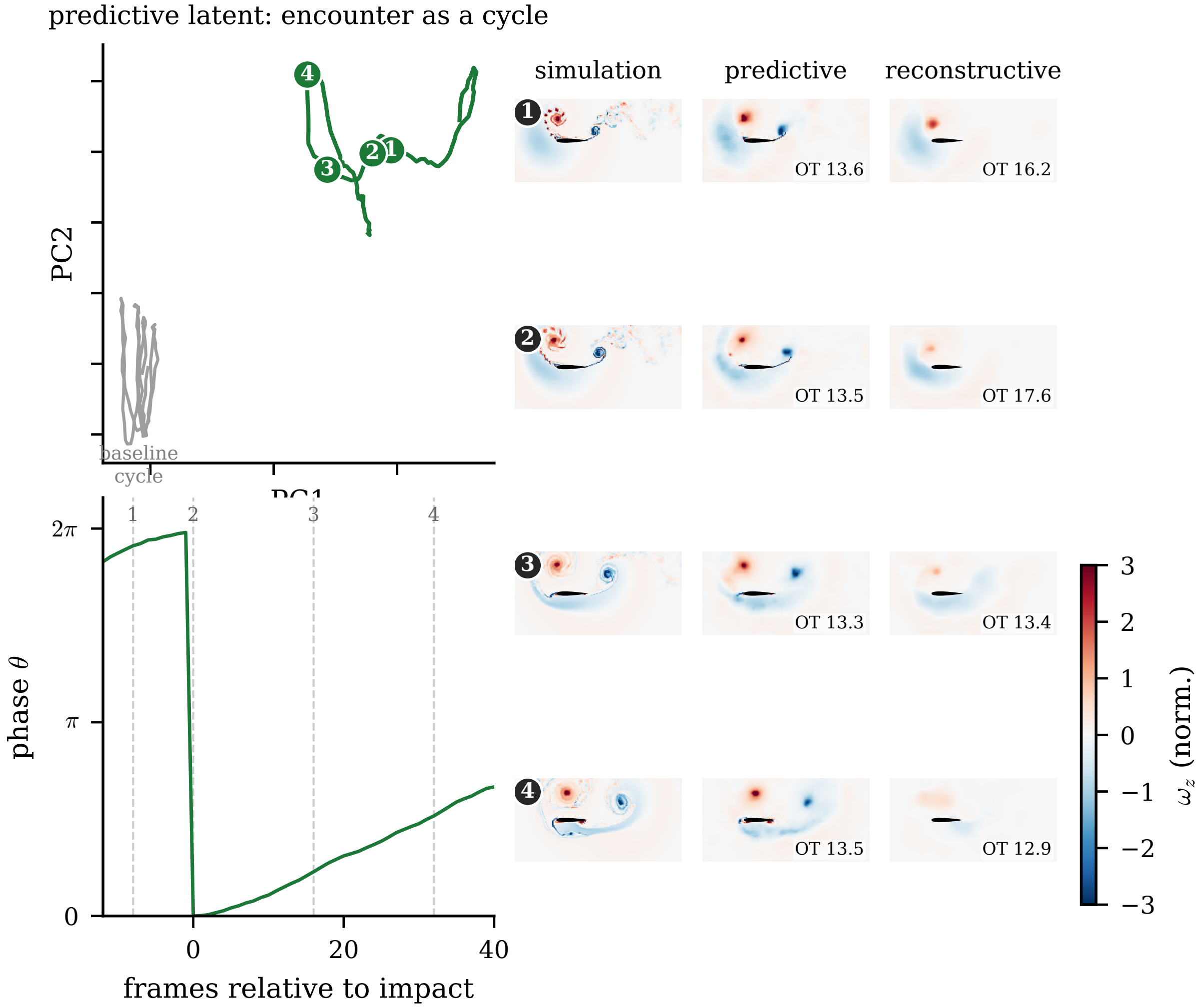

Figure 8: The gust encounter as a single closed cycle in the predictive latent, for a strong negative gust $(G, D, Y) = (-3.0, 1.5, -0.1)$. (a) The latent trajectory in the leading two baseline principal components departs the no-gust limit cycle (grey), executes the leading-edge-vortex growth and shedding, and returns; numbered glyphs mark the four stages and the arrow gives the direction of travel. (b) The orbit phase relative to the impact frame. (c–e) Mid-plane vorticity at the four stages for the simulation, the predictive decode, and the reconstructive decode; the predictive decode localises the impingement while the reconstructive decode collapses, with the optimal-transport field distance between each decode and the simulation annotated at each stage (largest for the reconstructive decode near impact). The stage glyphs are those of Figure 5.

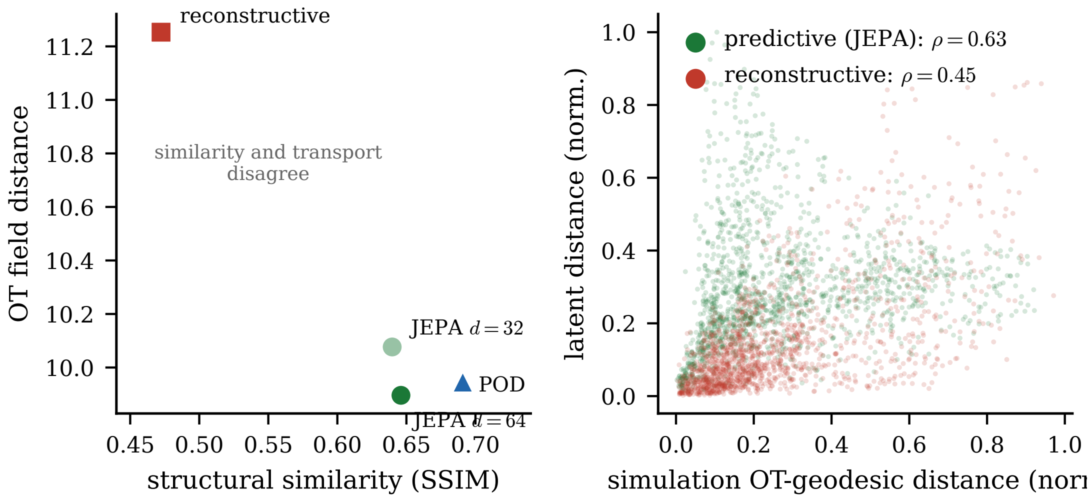

Figure 9: Optimal-transport reconstruction comparison (Tran et al., JFM 2026) on test_b. Unbalanced OT field distance by family at the impact frame, with the structural-similarity ranking shown alongside; the linear basis has the highest similarity but no transport advantage over the predictive decode, while the collapsed reconstructive field is correctly penalised. The Shepard panel shows the per-encounter alignment of latent distance with the simulation OT-geodesic, predictive versus reconstructive.

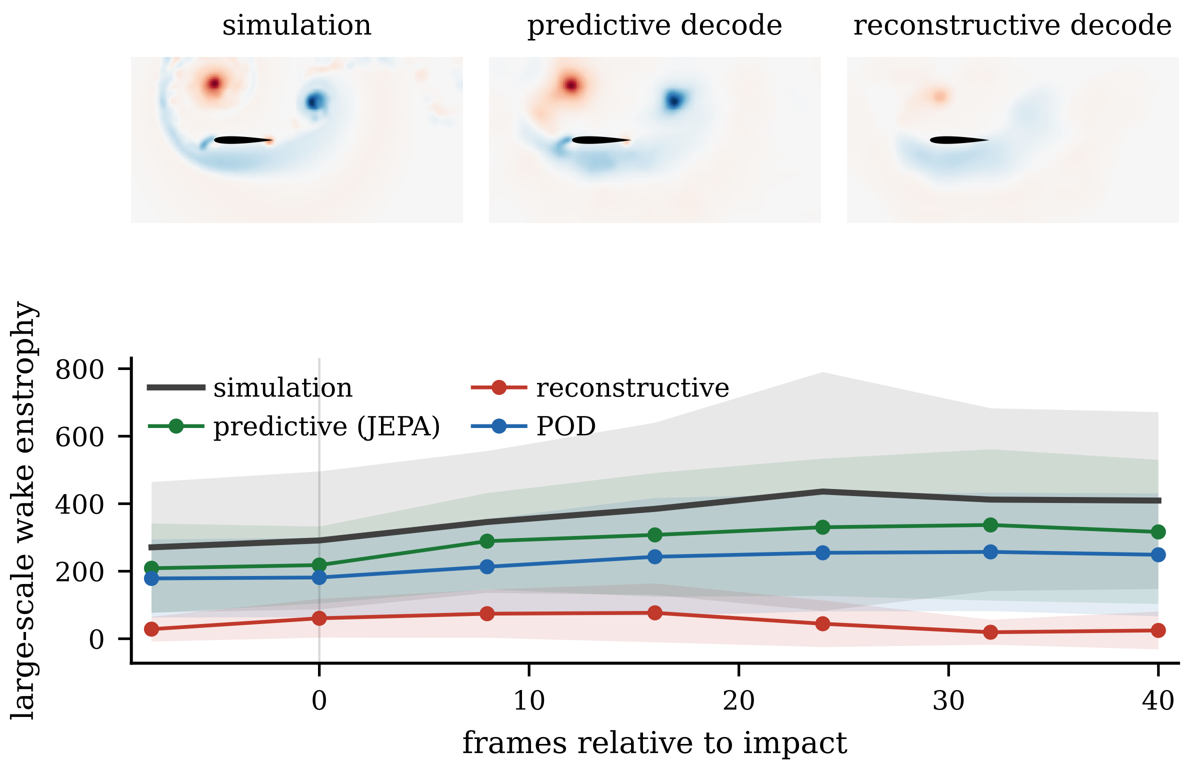

Figure 10: Gaussian scale decomposition ($\sigma/c = 0.05$) of the held-out reconstructions. Top: large-scale vorticity for the simulation, the predictive decode, and the reconstructive autoencoder at $H = 16$; the predictive latent retains the leading-edge vortex and shear layer the reconstructive one smooths away. Bottom: large-scale wake enstrophy through the staged encounter for the simulation, predictive, and reconstructive fields, on each split.

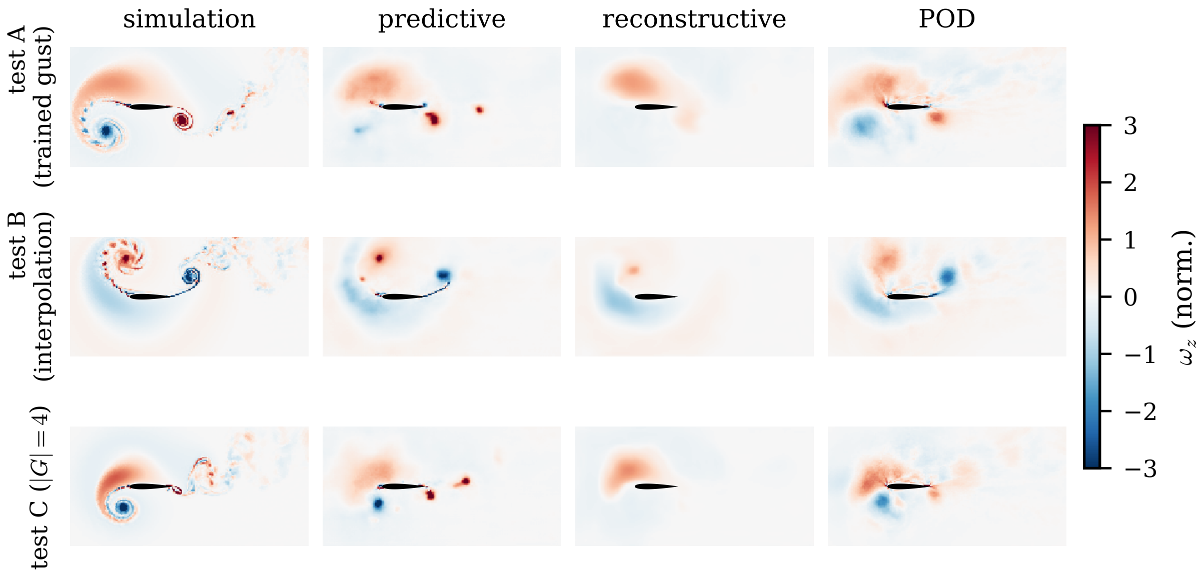

Figure 11: Decoded vorticity at the impact frame on held-out encounters from the three splits (rows: a trained-gust test_a encounter, an interpolation test_b encounter, and a $|G| = 4$ test_c encounter) for the simulation and the predictive, reconstructive, and POD decodes (columns). The predictive decoder is blurry by design but localises the impingement; the reconstructive autoencoder collapses toward a near-uniform field on the held-out interpolation condition, while the linear basis is amplitude-accurate. All families degrade in the $|G| = 4$ regime, reported as the observability boundary of §2.1 rather than as a generalisation claim. Field-space similarity is reframed by the optimal-transport metric of §4.6.

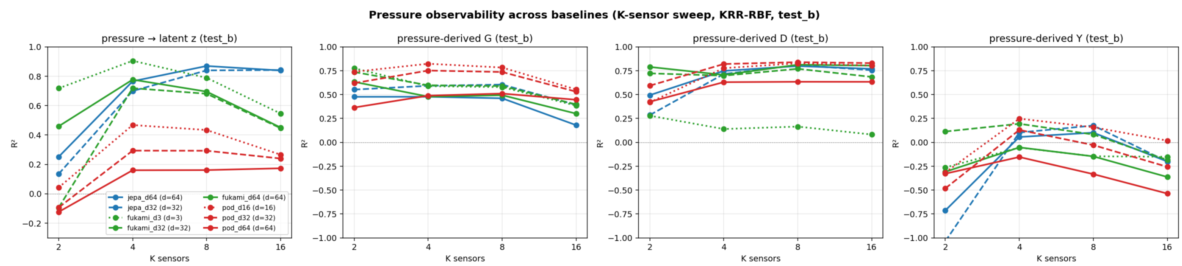

Figure 12: Pressure observability across encoder families on test_b, kernel-ridge estimator, versus sensor count K. Left to right: recovery of the latent state, and of the gust parameters G, D, and Y. The predictive (JEPA) latent is the most pressure-recoverable; G and D recover well, Y weakly. The out-of-distribution test_c counterpart (not shown) is negative for every family and is reported in the text.